Experiment – 7

Power Delay Profile, Delay Spread & RMS Delay Spread Analysis

Aim

To simulate and analyse the Power Delay Profile (PDP) of a multipath wireless channel, and to compute the Delay Spread, Mean Delay, and RMS Delay Spread using MATLAB, thereby characterising the time-dispersive nature of the channel.

Apparatus / Tools Required

Software: MATLAB R2024a | **Path Delays:** [0, 3, 5, 7] µs | **Path Gains:** [0.9, 0.7, 0.5, 0.3] | **Channel Model:** 4-tap multipath

Theory

In a wireless communication system, a transmitted signal arrives at the receiver via multiple paths due to reflection, diffraction, and scattering from obstacles. Each path introduces a different **time delay** and **attenuation**. The **Power Delay Profile (PDP)** captures how the received power is distributed across these multipath delays.

1. Power Delay Profile (PDP)

The PDP is a plot of normalised received power versus delay (in samples or microseconds). A Gaussian-shaped PDP (as simulated here) models environments where the power peaks at a central delay and tapers off symmetrically. It is mathematically expressed as:

$$P(\tau) = \exp(-(\tau - \tau_0)^2 / (2\sigma^2))$$

2. Delay Spread

The **Delay Spread** is the total range of delays over which significant multipath energy arrives. It is simply:

$$\text{Delay Spread} = \tau_{max} - \tau_{min}$$

3. Mean Delay (τ■)

The mean excess delay is the power-weighted average of all multipath delays:

$$\tau■ = \Sigma(\tau_k \cdot P_k) / \Sigma(P_k)$$

4. RMS Delay Spread (σ_τ)

The RMS delay spread is the square root of the second central moment of the PDP and is the most widely used metric for characterising multipath channel dispersion:

$$\sigma_\tau = \sqrt{[\ \Sigma((\tau_k - τ■)^2 \cdot P_k) / \Sigma(P_k)\ ]}$$

A large RMS delay spread implies severe inter-symbol interference (ISI) and necessitates equalisers or OFDM-based modulation in practical systems.

Procedure

1. Define path delays τ = [0, 3, 5, 7] µs and path gains P = [0.9, 0.7, 0.5, 0.3].
2. Generate a delay axis (0 to 100 samples) and construct a Gaussian PDP centred at sample 45 with a suitable standard deviation.
3. Normalise the PDP so that the peak power equals 1.
4. Plot the **Power Delay Profile**: Normalised Power vs. Delay (samples).
5. Compute **Delay Spread** = max(τ) – min(τ) = 7 – 0 = 7 µs.
6. Compute **Mean Delay**: τ■ = $\Sigma(\tau_k \cdot P_k) / \Sigma(P_k)$.
7. Compute **RMS Delay Spread**: $\sigma_\tau = \sqrt{[\ \Sigma((\tau_k - \tau■)^2 \cdot P_k) / \Sigma(P_k)\ ]}$.
8. Display numerical results using fprintf and publish the MATLAB script to PDF.

Observations

From the simulation (refer plots on subsequent pages):

- The Power Delay Profile follows a symmetric Gaussian shape, peaking (normalised power = 1.0) at approximately delay sample 45.
- The power rolls off smoothly from the peak, reaching ~0.2 at the edges (samples 0 and 100), indicating multipath energy spread.
- **Delay Spread** = 7.0000 µs — the total span of multipath arrivals.
- **Mean Delay** (τ■) = 2.7917 µs — the power-weighted centroid of the multipath delays.
- **RMS Delay Spread** (σ_τ) = 2.4829 µs — confirms significant temporal dispersion.
- Higher path gains at smaller delays (0.9 at 0 µs) pull the mean delay toward lower values.

Result

The MATLAB simulation successfully generated the Power Delay Profile for a 4-tap multipath channel. The computed parameters are:

- Delay Spread = **7.0000 µs**
- Mean Delay = **2.7917 µs**
- RMS Delay Spread = **2.4829 µs**

Conclusion

The experiment demonstrates the time-dispersive characteristics of a multipath wireless channel through PDP analysis. An RMS delay spread of ~2.48 µs indicates that the channel is highly dispersive — for any symbol rate whose symbol period is comparable to or less than this spread, ISI will be significant. In practice, systems like LTE and 5G NR use

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```
clc;
clear;
close all;
```

---------- PART 1: Power Delay Profile ----------

```
% Parameters
num_paths = 10;              % Number of multipath components
max_delay = 100;            % Maximum delay (in samples)
delay_spread = 20;          % Delay spread (std deviation)
power_dB = randn(1, num_paths); % Random power in dB

% Generate delay values
delay = randn(1, num_paths) * delay_spread + max_delay/2;

% Time axis
t = 0:max_delay;

% Initialize PDP
pdp = zeros(size(t));

% Generate PDP
for i = 1:num_paths
    pdp = pdp + 10^(power_dB(i)/10) * ...
        exp(-(t - delay(i)).^2 / (2*delay_spread^2));
end

% Normalize PDP
pdp = pdp / max(pdp);

% Plot PDP
figure;
plot(t, pdp, 'LineWidth', 2);
xlabel('Delay (samples)');
ylabel('Normalized Power');
title('Power Delay Profile');
grid on;
```

-------- PART 2: Delay Spread, Mean Delay, RMS Delay Spread --------

```
% Path delays (in microseconds)
path_delays = [0, 3, 5, 7];

% Path gains
path_gains = [0.9, 0.7, 0.5, 0.3];

% ---- Delay Spread ----
delay_spread_calc = max(path_delays) - min(path_delays);

% ---- Mean Delay ----
mean_delay = sum(path_delays .* path_gains) / sum(path_gains);

% ---- RMS Delay Spread ----
rms_delay_spread = sqrt( ...
    sum(((path_delays - mean_delay).^2) .* path_gains) ...
    / sum(path_gains) );

% Display results
fprintf('\n--- Delay Analysis Results ---\n');
fprintf('Delay Spread: %.4f microseconds\n', delay_spread_calc);
fprintf('Mean Delay: %.4f microseconds\n', mean_delay);
fprintf('RMS Delay Spread: %.4f microseconds\n', rms_delay_spread);
```

```
--- Delay Analysis Results ---
Delay Spread: 7.0000 microseconds
Mean Delay: 2.7917 microseconds
RMS Delay Spread: 2.4829 microseconds
```